

A Novel Compact and Portable Linear Accelerator Electron Beam System

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Abstract—A lightweight, portable 15 MeV electron beam linear accelerator (LINAC) system has been developed for research applications. Our approach uses a commercial, off-the-shelf magnetron to drive an S-band standing-wave accelerator. The magnetron is driven by a Type-E pulse forming network (PFN), the modulator, switched with a thyratron. Within the modulator macropulse, short (<100 ns), high-frequency (up to 3500 Hz) electron beam pulses are produced by gating the electron current from a triode electron gun. The entire system is contained within approximately 2 m³, weighs 250 kg, and can be disassembled into two person-portable components, each weighing 50 kg or less.

Index Terms—Electron beams, electron linear accelerators (LINACs).

I. INTRODUCTION

THE Directed Energy Directorate at the Air Force Research Laboratory (AFRL) has had a long-standing interest in electron beam sources of various types for defense, security, and basic research applications. These interests suggested the development of a compact, portable, Ampere-class electron beam linear accelerator (LINAC) in the sub-20 MeV energy range. Electron beams in this parameter space have a wide range of applications, including cargo inspection [1], [2], [3]. The additional challenges of small size and portability enabled the reuse of a small, existing shield vault for initial testing; provided flexibility for off-site testing for future AFRL and joint service projects; and gave

Parameters	Objective Range
Beam Kinetic Energy	12-18 MeV
Beam Spot Size (diameter)	<2 cm at 20 cm from the exit window
Pulse Width	20-40 ns
Pulse Separation	100-300 ns (adjustable)
Instantaneous Pulse Current	0.5-2 A (adjustable)
Maximum Pulse Repetition Rate	3500 Hz

an opportunity to explore size, weight, and power (SWaP)-constrained technologies for prime power, pulsed power, and RF generation, which are of interest for other AFRL missions.

Toward this end, a lightweight, portable electron accelerator system was sought, having the specifications summarized in Table I.

In addition to the short-pulse mode of operation emphasized in Table I, a conventional, constant-current mode of operation was also desired. Finally, it was necessary that the source be lightweight (approximately 250 kg), portable, and designed so that it could be disassembled into components/subsystems, each weighing no more than 50 kg. Because of facility restrictions, the width of the accelerator had to be less than 60 cm, the length could be no more than 1.2 m, and the height had to be less than 1.5 m.

II. PRELIMINARY DESIGN CONSIDERATIONS

The kinetic energy, size, and implied gradient requirements are typical of an electron beam LINAC, implying that standard design approaches could be used [4], [5]. Such a LINAC typically operates in a pulsed mode in which the microwave source supplies macropulses at a certain pulse repetition frequency (PRF). Application of a macropulse to the LINAC produces a stream of electron pulses (micropulses) for each input macropulse at the frequency f of the microwave source.

The efficiency of a $\pi/2$ -mode standing-wave LINAC can be written as follows [6]:

$$\eta = [1 + E_g/(ZI)]^{-1}. \quad (1)$$

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E_g is the overall accelerating gradient, Z is the shunt impedance per unit length, and I is the average macropulse beam current. By definition $\eta = V_{\text{beam}} I / P$, where V_{beam} is the beam voltage and P is the peak power of the microwave source. Assuming a LINAC length of 1 m and $Z = 75 \text{ M}\Omega/\text{m}$ (characteristic of good S-band accelerating structures), the macropulse beam current (in A) can be written in terms of the beam voltage (in MV), and the source power (in MW) as follows:

$$I = P / V_{\text{beam}} - V_{\text{beam}} / 75 \text{ M}\Omega. \quad (2)$$

The product of the macropulse current and the duty cycle (pulse duration multiplied by the PRF) equals the average current. As an example, for a macropulse current of 0.13 A, the source power required to deliver 15 MeV ($V_{\text{beam}} = 15 \text{ MV}$) would be $\sim 5 \text{ MW}$. At a duty cycle of 0.1%, the average source power would be 5 kW, and the LINAC efficiency would be about 40%.

Given these arguments and the restrictive weight limitations, we selected the MG6028 Precision Tuned Magnetron from E2V. It has a peak power capability of 5.5 MW at a duty cycle of 0.08%. Its nominal center frequency is 2.856 GHz, with a tuning range of 10 MHz. The tube itself only weighs 8 kg, but its integral electromagnet and output section weigh 50 kg. To drive this tube, the high-voltage Type-E pulse forming network (PFN) modulator must supply 50 kV, $3.5 \mu\text{s}$ current pulses into 200Ω . Because of the LINAC's short length, the effective gradient of the accelerator (including transit time effects) exceeded $\sim 13.6 \text{ MeV/m}$ to achieve a 15 MeV electron beam, and the electron gun supplying the electrons needed to attach directly to the first cavity of the LINAC.

For reduced size and system simplicity, it was convenient to operate the LINAC electron gun at the same 50 kV voltage as the microwave tube, but generating the short pulses required a triode configuration in which electrons are emitted by applying a voltage to the gap between the cathode and a grid. The EIMAC Y796 cathode grid assembly meets this requirement. The EIMAC Y796 cathode area is $\sim 2 \text{ cm}^2$, and the maximum perveance is about $0.2 \mu\text{A}/\text{V}^{3/2}$. Thus, the cathode-grid voltage required for gating the gun is only a few tens of volts.

The beam spot size is limited by scattering in the thin exit window. A crude approximation for the mean square scattering angle is given in the following equation [7]:

$$\langle \theta^2 \rangle = ax(Z/E)^2. \quad (3)$$

Here, x is the thickness of the scattering medium in cm, Z is the atomic number, and E is the electron energy in MeV. For a titanium window, this becomes

$$\langle \theta^2 \rangle = 80x/E^2. \quad (4)$$

To meet the beam spot requirement in Table I, the electron beam in the LINAC needed to be no larger than 1 cm in diameter and the titanium window thickness no more than $50 \mu\text{m}$.

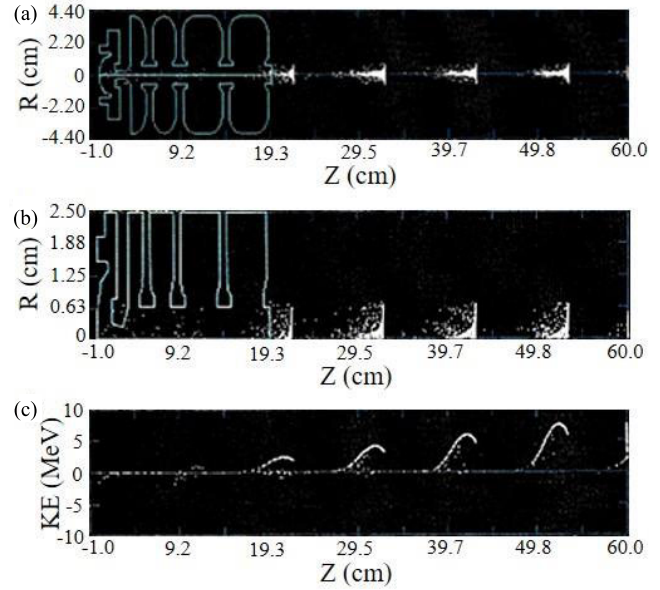


Fig. 1. (a) TWODET simulation for a 10-MeV version of the S-band LINAC, showing electron bunching and acceleration. Only the first four accelerating cells are shown. (b) Zoomed-in view of geometry and bunch radius. (c) Particle kinetic energy versus distance.

III. COMPONENTS AND SUBSYSTEMS

A. LINAC Structure and Electron Gun

For the 2.856 GHz frequency, a good LINAC design strategy was readily available [7]. The LINAC consists of a biperiodic $\pi/2$ structure with axial coupling. There are two symmetric coupling slots per cavity wall, and the two slots are rotated by 90° on either side of the coupling cavity to reduce the coupling between adjacent accelerating cells. (The coupling slots on opposing accelerating cavity walls are, however, not rotated.) To generate a $\sim 15 \text{ MeV}$ electron beam, the LINAC was chosen to consist of 17 full-length accelerating cells, one elongated half-cell (attached to the gun), and one transition cell; the total length of the accelerating structure was 97 cm. Results from Poisson SUPERFISH [8] software show that the ratio of the peak field stress to the average field was 3.10 in the full accelerating cells. Because of these accelerating gradients, the transit time factor for the electrons was 0.806, and the associated bore radius was 0.65 cm. The shunt impedance, Z , is larger than expected for an S-band accelerator. This is due to the idealizations made by the SUPERFISH model. This calculation was also done in 2-D, which ignores the 3-D losses associated with the coupling slots. The quality factor, shunt impedance, transit time, and peak surface electric and magnetic fields normalized by E_z averaged across the cell E_0 are shown in Table II. As an example of the expected beam behavior, an output frame of a TWODET (a fixed-frame version of PARMELA [9]) simulation for a 10 MeV version of the LINAC is shown in Fig. 1. Note that the beam diameter is approximately 1 cm without a magnetic focusing coil.

The aperture of the coupling iris from the waveguide feed into the LINAC is usually chosen to minimize reflections, but this choice is complicated by the need to operate in a short, repetitive pulse, high-current mode, as well as in a lower

TABLE II
LINAC SUPERFISH DESIGN PARAMETERS

Linac Parameters	Values
Quality Factor Q_0	17,425
Shunt Impedance Z	121 M Ω /m
Transit Time Factor	0.806
$E_{\max}/E_0$	3.10
$H_{\max}/E_0$	2.05 kA/MV

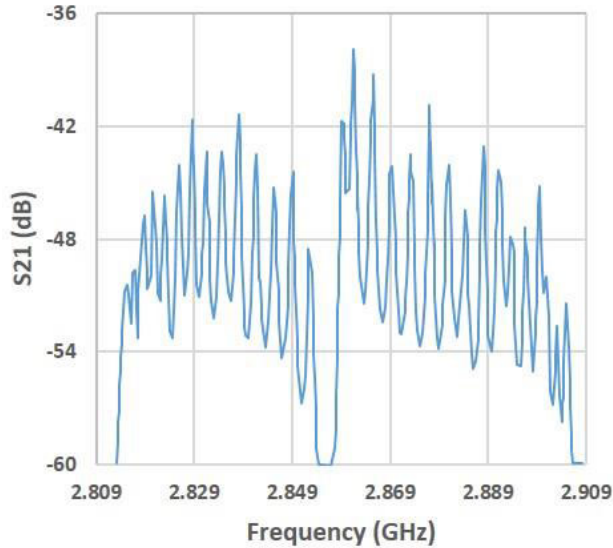


Fig. 2. Final cold test of the brazed LINAC structure. The dip in S21 at approximately 2.858 GHz shows the difference in resonances of the coupling and accelerating cavities to be approximately 2 MHz. The cold-start frequency of the system is the frequency of the first resonance above the stopband, in this case 2.861 GHz.

steady-current mode. A further complication is that the pulse from the magnetron is only $\sim 3 \mu\text{s}$ in duration. We finally selected a compromise design goal of 2.5 for the coupling coefficient of the input iris.

With these considerations in mind, copper accelerating cells were fabricated, and cold tests were performed to refine the designs to achieve the desired resonant frequency and coupling coefficients. The cells were then brazed to form the complete accelerating structure. The final cold test of the structure is shown in Fig. 2. In comparison with the pre-braze results the resonant frequency of the accelerating cavities increased to ~ 2.859 GHz, and that of the coupling cavities changed to ~ 2.857 GHz, implying a stopband of ~ 2 MHz. From this, the cold-start frequency (the frequency of the first resonance above the stopband) was estimated to be 2.861 GHz. Since these frequencies will decrease with heating as the LINAC is operated, these results were judged to be acceptable. The coupling coefficient from the accelerating cell to the coupling cell was about 2.8%, and the measured value of the coupling of the waveguide iris, $\beta_0 = 2.4$, was also judged to be acceptable.

For the short pulse mode, a macropulse duty cycle of 0.08% implies a maximum PRF of 267 Hz for $3 \mu\text{s}$ voltage pulses from the PFN. To meet the electron beam PRF of 3500 Hz requires 13 pulses per macropulse at a frequency of 4.4 MHz. Assuming 30 ns electron beam pulses, the peak current during

a short pulse was ~ 0.98 A. Assuming a capture efficiency of 45%, the implied perveance was $0.19 \mu\text{A}/\text{V}^{3/2}$. During the design process, the gun region (Pierce geometry) was designed using the EGUN code [10] to have a perveance of $0.2 \mu\text{A}/\text{V}^{3/2}$, corresponding to the EIMAC Y796.

To generate the short, high-frequency grid pulses, custom-designed printed circuit board (PCB) drive circuits using Directed Energy, Inc. (DEI) fast MOSFETs were used. A custom-designed fiber optic driver circuit was driven by an output from a Quantum pulse generator (for timing purposes) into an Avago fiber optic receiver mounted on the control grid PCB driver. Using this setup, the cathode-grid assembly was operated at a 4 MHz rate over a $3 \mu\text{s}$ window at a simulated modulator pulse rate of 250 Hz.

A final picture of the integrated gun and LINAC sub-assembly mounted to its support plate is shown in Fig. 3(a). The entire length is approximately 120 cm from the input flange of the gun assembly to the titanium exit window. Note also that the waveguide input coupler is located at the 14th accelerator cell. This location provided adequate room for integration with the RF subassembly on a wheeled tripod base.

B. RF Subassembly

The RF subsystem consists of the E2V magnetron and its associated electromagnet, all the waveguide components, including E- and H-plane sweeps, a three-port isolator with integral water load (Model 102 647 from EM Design), gas ports, and the ceramic RF window at the LINAC input coupler port. A picture of the RF system is given in Fig. 3(b). Tuning of the MG6028 magnetron is accomplished using a dc power supply. The tube and electromagnet are water cooled, and the output waveguide must be pressurized. An important specification for the modulator is a voltage rise rate of $\sim 125 \text{ kV}/\mu\text{s}$ to ensure that the magnetron starts oscillating in the correct (π) mode. The electromagnet (E2V Model MG6030) is the single heaviest component in the entire LINAC system at 50 kg.

C. High-Voltage Modulator

The high-voltage modulator uses a standard line-type configuration. An ALE 1202L 30 kV power supply (manufactured by TDK-Lambda) charges the capacitors of a Type-E PFN. The PFN is switched into a 3.6:1 voltage step-up transformer (manufactured by Stangenes Industries) using an HY-5 hydrogen thyatron. The modulator components are housed in a commercial off-the-shelf frame having a rectangular volume of approximately 0.25 m^3 . The physical layout of the modulator components is shown in Fig. 4.

D. Auxiliary Subsystems

Several auxiliary subsystems are necessary for the proper functioning of the LINAC system, including vacuum pumps, an SF_6 system for pressurizing the waveguide to 20 psig (239 kPa), a small Paxton air blower for cooling the exit window, and a water cooling system for heat removal and temperature control of the RF system and LINAC. Three vac-ion pumps are used; one 20 L/s pump is attached near the exit window,

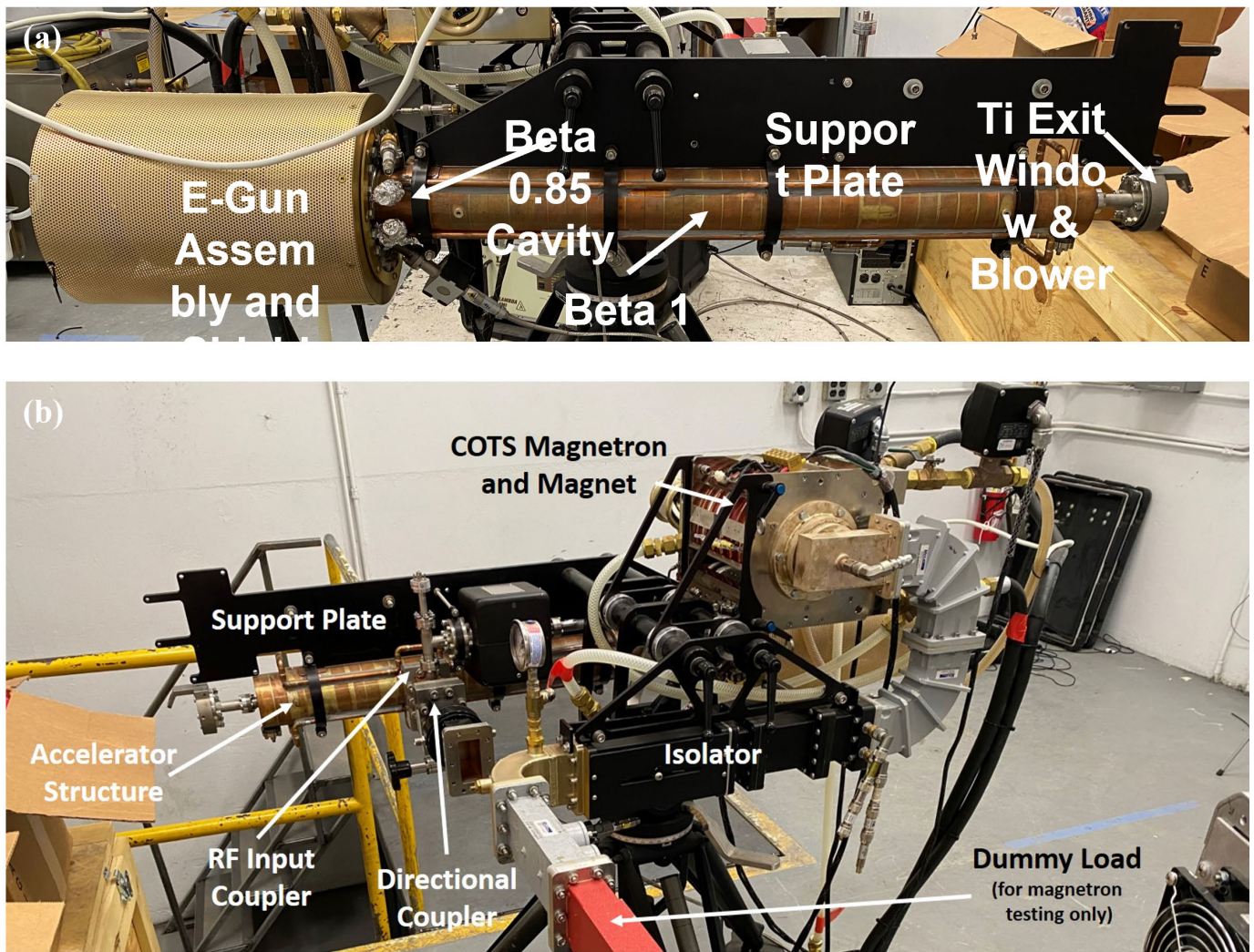


Fig. 3. SPEBS accelerator. (a) Accelerating structure. The copper LINAC section is 97 cm long with a diameter of 9.4 cm. (b) RF system. Note that the system is in a configuration to test the magnetron. During actual operation, the gold elbow on the output of the isolator would be facing the directional coupler. Not shown: prime and pulsed power cabinet and cooling and control cabinet.

another is attached near the LINAC input coupler port, and a small 2 L/s pump is attached near the electron gun region. With all-metal seals and valves, the vacuum pressure in the LINAC (when cold) will easily descend into the 10^{-9} torr region.

The overall requirements for the important LINAC system components requiring water cooling are summarized in Table III. To meet these demands, a two-chiller approach was adopted. Cooling for the larger system components (high-voltage power supply (HVPS), magnetron, and electromagnet) is provided by a two-ton Advantage chiller, and the temperature stability of the LINAC is provided by a small Lytron chiller.

E. Controls

The control system consists of two embedded controllers, a Quantum delay generator, a laptop computer (tablet), and several peripheral equipment items. The system graphical user interface (GUI), located on the tablet, allows the user to set system parameters, monitor system conditions and safe arm,

TABLE III
COOLING REQUIREMENTS FOR SOME IMPORTANT SYSTEM COMPONENTS

Device	Flow (L/min)	Max T_{in} (°C)	Max T_{out} (°C)	Q (kW)
HVPS	8	35	37.5	1.3
Magnetron	27	67.8	70	4.2
Electromagnet	5	40	44.7	1.5
LINAC	15	24.2	26	1.9
Totals	55			8.9

and fire the system. The main embedded controller, an Arm9 PC-104 mounted in the control chassis, contains analog input–output (I/O), digital I/O, serial communications, and Ethernet communications. It controls and monitors peripheral equipment and takes appropriate action if a fault condition is detected. The peripheral equipment includes the water chillers, temperature and flow monitors, a programmable integrated chip (PIC) that controls system timing, the magnetron tuner, the vac-ion pumps, the blower for exit window cooling,

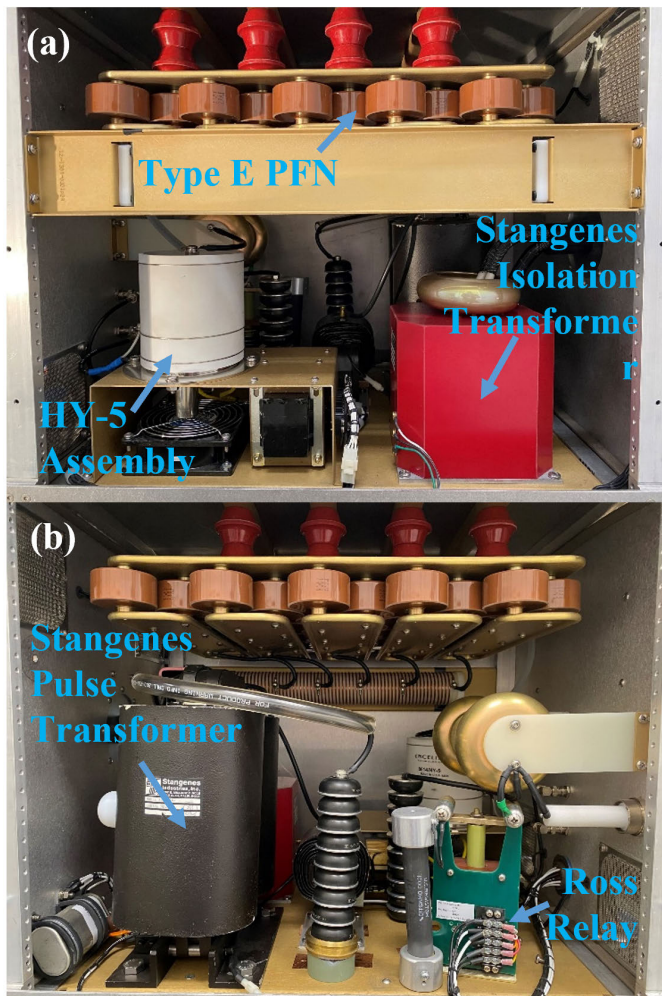


Fig. 4. Layout of the as-built HV modulator, front (a) and back (b). The modulator compartment is 61 cm across and 53 cm high.

and the electromagnet power supply. The PIC generates a heartbeat pulse that controls the firing of the thyatron to produce a high-voltage pulse. It also monitors the reflected power from the LINAC and reports these data to the main controller. This information is used to tune the magnetron to the LINAC frequency as the temperature changes. The Quantum delay generator provides timing pulses (referenced to the PIC heartbeat) for driving the grid of the electron gun for short pulse operation.

IV. SYSTEM PERFORMANCE

Preliminary tests were performed on the short-pulse electron beam system (SPEBS) to determine the emission characteristics of the electron gun, the short-pulse grid drive behavior, magnetron power and tuning curves, and PFN performance. The LINAC structure was operated at low magnetron power and pulse repetition frequencies and without electron gun current to ensure satisfactory performance of the controls and to perform some RF conditioning of the structure. It was verified that the cold-start frequency of the evacuated LINAC was about 2.861 GHz. Based on the reflected power signals, the LINAC fill time was estimated to be of the order of $1 \mu\text{s}$, as expected.

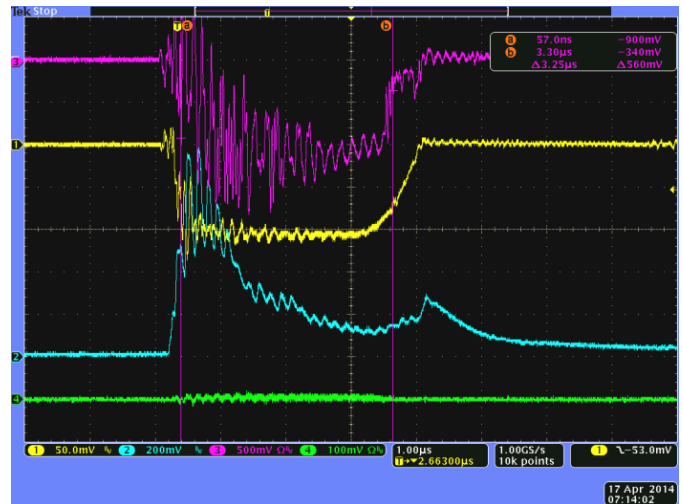


Fig. 5. Comparison of magnetron output power (yellow), collected beam current (magenta), reflected power (blue), and Intermediate Frequency measurement (green) for modest beam loading conditions (0.22 A). The grid drive was ~ 20 V, and the PFN voltage was 25 kV.

After the preliminary tests, full system operation was initiated, starting at low PFN charge voltages and low pulse repetition frequencies. Typical data for operating in the constant-current mode are shown in Fig. 5. The beam exiting the LINAC was sensed by a Faraday cup made of 4.1 cm of aluminum (sufficient to stop electrons with energies up to 20 MeV).

The beam current amplitude depends on the electron gun voltage, the amplitude of the grid voltage in the electron gun, and to a lesser extent, the fields in the LINAC. For modestly high grid voltages (~ 50 V), the beam current is space-charge limited by the grid-cathode gap voltage, and therefore varies with PFN voltage. Apart from this variation, control of the beam current is largely dependent on the amplitude of the grid pulse because the magnetron must be operated at voltages high enough to generate greater than 5 MW RF pulses. For example, running the grid drive at ~ 50 V with a PFN charge of 25 kV will generate a steady LINAC current pulse of ~ 0.36 A; however, this excessive beam current overloads the LINAC such that the electromagnetic fields are not sufficient to capture and bunch the beam at late times in the pulse.

A calibrated magnetic spectrometer was not available for detailed kinetic energy measurements. Consequently, only crude beam energy determinations were made using range-energy data and beam spot size measurements. For a 25 kV PFN charge and 20 V on the grid (beam current of ~ 0.14 A), electron current was clearly detected through 2.8 cm of aluminum (Fig. 6), but not through 4.1 cm. From the tables, the kinetic energy is, therefore, in the range of 13.2–20.8 MeV. In addition, the depth of Lichtenberg dendrites in acrylic (density 1.15 g/cm^3) was approximately 9 cm, implying a beam energy upper limit of about 18.5 MeV. Spot size measurements at 20 cm from the exit window were also consistent with a maximum beam energy of approximately 18 MeV. In comparison, for an iris coupling coefficient of 2.4 and 5.5 MW of power, the estimated LINAC load line is Beam Energy (MeV) = $19.4 - 24.3I$ (A), and the predicted electron energy is 16 MeV.

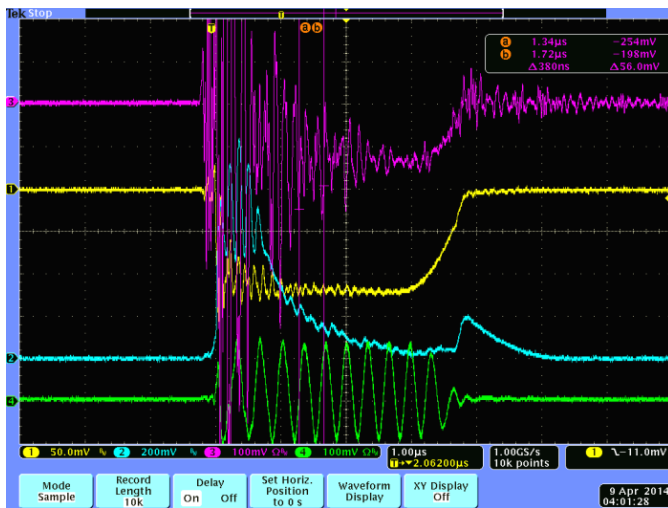


Fig. 6. Beam current (magenta trace) detected through 2.8 cm of aluminum.

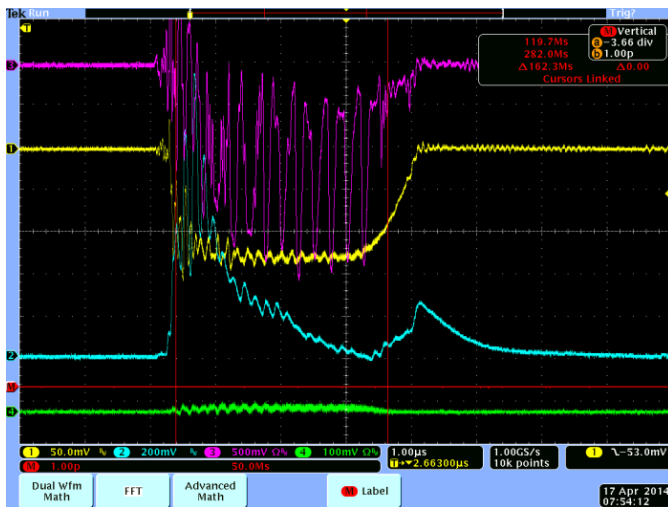


Fig. 7. Final pulse of a test that fired a total of 2000 pulses showing twelve 30 ns pulses at 4 MHz on the electron gun grid voltage.

In the second mode of operation, short pulses are applied to the grid of the electron gun at frequencies up to 4 MHz. The final pulse of a 20 s, 100 Hz burst is shown in Fig. 7, with 26 kV on the PFN and 50 V on the grid drive. There are 12 short pulses in the 3 μ s macropulse. At a PFN voltage of 28 kV, and 100 V on the grid, peak currents of ~ 0.9 A were observed during 40 ns short pulses. The beam energy will typically decrease by a few percent over the duration of a short pulse due to the depletion of the energy stored in the LINAC cavities.

As the modulator PRF is increased above 10 Hz, heating of the magnetron causes its frequency to decrease, and it is necessary to increase the tuner voltage to maintain a constant frequency. A time- and frequency-dependent tuning algorithm was implemented in software to mitigate this effect. With this algorithm, the system has operated at PRFs as high as 250 Hz.

Beam spot size measurements were performed to examine beam expansion resulting from scattering in the exit window. Images taken using B3 radiochromic film at 10, 20, and 30 cm from the window are shown in Fig. 8. All of these

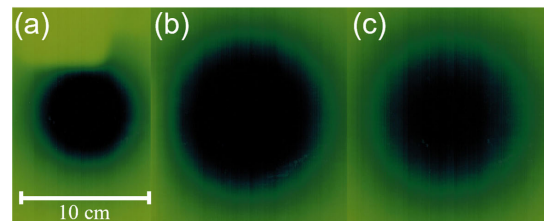


Fig. 8. Single macropulse beam images taken at (a) 10 cm, (b) 20 cm, and (c) 30 cm from the beam window.

were taken in short-pulse mode with eleven 50 ns pulses in the macropulse. Note that the beam expands approximately linearly from a little less than 1 cm in diameter at 10 cm to about 2.6 cm at 30 cm. There was no loss of beam current over these distances.

V. CONCLUSION

This work demonstrates the design and construction of a compact, portable S-band RF LINAC weighing less than 250 kg with a total system volume of approximately 2 m³. The system was shown to produce a beam of approximately 15 MeV and up to 0.9 A peak current. This technology has significant applications in research and development but can also find utility in port inspections and for radiation hardness testing against fixed targets. This work significantly broadens the potential applications of charged particle beam technology.

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